Modern C++17 Smart Pointers & Memory

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DPU Course Material

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Unique Pointer & Move Semantics

- • Zero-overhead exclusive resource ownership
- • Move-only semantics prevent accidental aliasing
- • Destructor guarantees automatic heap

```cpp
// CPP
#include <iostream>
#include <memory>

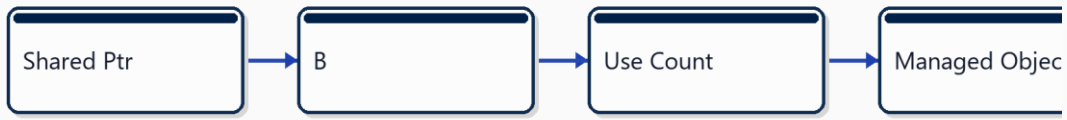
struct Buffer {
    int id;
    Buffer(int i) : id(i) {}
};

std::unique_ptr<Buffer> make_buffer(int id) {
    auto buf = std::make_unique<Buffer>(id);
    return buf; // Move semantics applied
}
```


cleanup

Shared Pointer Control Block Architecture

- • Atomic reference counting across threads
- • Weak pointer breaks cyclic reference leaks
- • Control block allocated alongside payload



Modern C++ Memory Safety Rules

- • Strictly avoid raw pointer new and delete calls
- • Use std::make_shared for unified cache locality
- • Break circular ownership graphs with weak_ptr
- • Pass by const reference to avoid ref-count bumps
- *Exercise Reference: exercise_3_cpp_raii*